An Integrated Courier Services Application: A New User Experience

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***Abstract*—There are numerous courier services available locally or worldwide. However, the most services only applicable at their respective company's whereby the pricing information is often hidden and customers usually need to call the company or visit their respective website or application to get quote, then make a price comparison with other company's courier services quotes for them to get their preferable quotation. This could be very time consuming mainly due to the inflexibility services whilst with certain terms and conditions apply only customers able to enjoy the door-to-door delivery service. Hence, the main objective of this project is to integrate a few of the courier services price and information in one stop whilst providing a low-cost peer-to-peer service to assist customer in delivering their parcel. The application will be able to generate the quotes of all the selected courier services in one application to ease the customer and enable a price comparison for each of the selected courier services company. The usability study was conducted and analysis of respondents' feedbacks shows positive interest of having this application and would opt to use the low-cost peer-to-peer services. With Parcel2Go Mobile Application, the problem of inconveniency of the customer and the inflexibility of services that have in the current courier services will be solved as this application will ease the people who wants to deliver their parcels.**

Keywords—Courier Services, Price Comparison, Low-Cost Peer-To-Peer Service, Mobile Application.

I. INTRODUCTION

The population of Malaysia in the year between 2014 to 2016 has approximately 31.7 million and the annual growth rate between 2015-2016 is 1.5% [1]. Among its evolving population, constant movement of people and goods from one location to another is expected to occur due to certain cause or situation. If an individual who does not possess their own transportation (vehicles) to travel from one place to an-other, it can be a quite stressful process for them [2]. This barrier eventually happens when an individual is initiating to perform the action of delivering parcels and goods. Thus, door to door delivery service can be invaluable help for the individual [3].

Courier services can be defined as a contracted company whose responsibility as an outsourcing medium in assisting the role of transporting the goods of a customer from one geolocation to another [4]. Generally, the goods are related to the delivery of documents, packages, printed materials, parcels and letters [5]. Typically, the postage transportation process consists of mail collection, input sorting, organize the movement of mail (global area transportation), output sorting and mail distribution. Thus, it is better for ICT to assist the complexity of the process to make sure that the correct mail can be received by the right customer on time.

Nevertheless, an average person or online seller would not easily access to the courier services due to their inflexibility in services. This is because it requires a company registration, office address and a monthly committed volume for them to enjoy the privilege of courier services. This has directly placed numerous online sellers in a dilemma situation as the high shipping cost of courier services would turn customers away, whereas normal mail services would take a very long time to deliver the pack-ages to the buyers.

Besides, customers need to run around to quote rate such as to make a call to the courier service company or visit their respective website to get quotation. With that, a price comparison could be done with other company's courier services rate to get their preferable rate. This is because different company courier services will have a different rate thus it is time consuming for the customer to check one by one.

In addition, customers tend to feel troublesome to travel to the post or courier office in purpose to post their documents if the distances are far apart yet not owning any transportation. The door-to-door delivery service that provided by courier services could only be enjoyed with certain terms and conditions apply or additional charges will be charged to the customer.

Therefore, this paper presents a study that integrate the selected courier service providers in one stop to save user's time in terms of price comparison while develop a low-cost peer-to-peer courier service for the users. The scope of the study does not only limit itself to the concept of integrating the quotation of the courier services but also matching up people who wants to deliver their parcels and people willing to help them and earn while travelling.

II. LITERATURE REVIEW

A courier company is one which delivers messages, packages and mail is recognized got their speed, security, tracking service and specialization [6]. Today, with the advancement of technology by ensuring that the customer will receive the right mail on time, the Information Communication Technology (ICT) has been implemented in the postal and

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courier services process to improve the delivery process efficiently.

Courier services operate on all scales, which are from town to town, district delivery services and worldwide service. Among the courier companies consist of DHL, GDEX, UPS, FEDEX etc. The delivery pricing must be competitive and adaptable where it is also easy to use and reasonable rates to the value of goods, weight of goods, in addition the distance of delivery [4]. A right pricing plan shall be automatically provided to the confidence of the customer [7]. Therefore, courier services to-day could be also defined in such a way that the trust between courier services as a business partner that logistic services e- commerce by improvising the performance of e-business from the business side with delivering important parcel.

For Malaysia, to develop courier service management more competitively, there still needs to do a continuous learning and research. It is to understand the type of products that are profitable in addition not handling with the continuous modification in the nature of business based on the e-commerce environment irrespective of their scopes deal with challenges in facing with the technology advancement from a business partner [2]. Hence, a constant change is need to be made to gain competitive advantage in e-commerce business environment, services and operations, and customer where the only determinant of the courier service industry is no longer only the price [8].

The relationship between both the courier and logistic services companies have some symbiosis relations where the courier service providers and customer such e-commerce enterprise will use the logistic service repetitively [4]. The logistic management strive to boost competitive enhancement by carefully assimilating the core functions inside the organization to other business. Whilst with the outside operations of supplier activities, customer prerequisite and other network members should be connected effectively with the business partner. To express it in another term, the outcome of the customer perspectives will be resulting a good organizational performance with business partners like courier services companies as a CPL to e-commerce enterprise [9].

Logistics are not limited to manufacturing operations solely and it is applicable to all types of organization for instance to serve e-commerce enterprises [10]. Certainly, all the enterprises particularly in the retail industry, large organization, small and medium enterprises could relate to the logistic management [11].

Apart from that, the collaboration in the courier service offers multi-innovative features and require a long time to build a friendly alliance in courier service and business partner [4]. Hence, the need of flexibility in services to meet the wants and demands of e-commerce enterprise companies is vital for the courier service providers. From here we can observe that how the performances of e-commerce enterprises could positively be affected by the courier services.

New demands in the courier services have been formed due to the rapid development of ecommerce enterprise in network. For instance, e-business's customer who has demanded the different which simultaneously provide a service which is value-added in courier services are increasing. Formerly, the conventional method of its standing made it hard to seek for services to meet these new needs. Thus, the development of e-commerce enterprise which can improvise the position of courier service providers in the forthcoming as relevance is analysed, where it is also impactful for the economy development in Malaysia especially.

In the era technology of the desktop PC the web browser was the main method of access to these resources; however, everything has changed due to the explosion of smartphones and tablets. In spite of every handset being equipped with a mobile browser, users are instead choosing to use apps for various common functions such as downloading and consuming music and videos, and reading news [12]. According to Angeles [13], the prime benefit is that the mobile users are able to access to the regular website. It could obtain all the same elements as what they contain in the regular version of the website for instance its design and content of the pages and it features a mobile friendly layout that offers improved readability and functionality when viewed on a smartphone or tablet. There are quite a number of similar mobile applications exist in the Google Play and AppStore but only three standard applications namely ezyCourier, FastFast and eCourierz. Although these applications have similarities, they do not have a functionality whereby both the price comparison of domestic courier services and a peer-to-peer delivery service is also provided in one-stop application

III. METHODODLOGY

A. *System Architecture*

There is an important task is to identify through the experience of the ecommerce and an average person who will be involved with the study that wants to send a parcel to another party. Then, the development will integrate the price rating of certain domestic courier services in one stop and meanwhile provide a low-cost peer-2-peer courier service to ease both online seller and an average person.

There will be a two-way communication which involves both parties who is the customer and the Par-cel2Go personnel as shown in Fig. 1. Whenever customer wants to deliver their parcel, they will first need to send a delivery request and it will be process through the web-server and store in the database before the Parcel2Go personnel could receive the customer's request. After the personnel had received the request from customer, the personnel will need to give a confirmation to the customer. Confirmation is then again go back through the web server and stored it into the database before customer received the confirmation.

There are two type of users which will be accessing the difference of contents and functionalities. For instance, the customer can access the content of sending a delivery request whilst the Parcel2Go personnel have the functionality of viewing a list of delivery request whether to accept or reject.

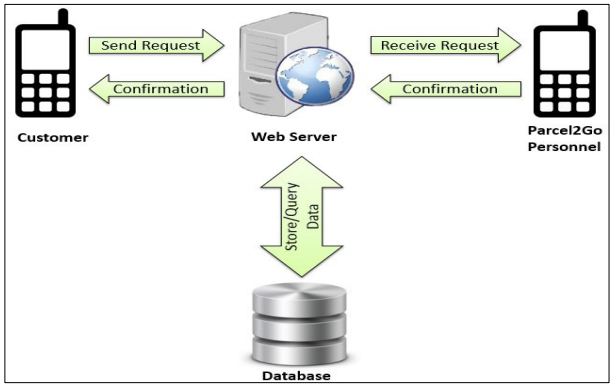


Fig.1. System Architecture for Parcel2Go Mobile Application

B. System Design

Despite the fact there is a similar existing application that already exist in market, however there are still certain loopholes whereby the online business seller and courier service user find the inconveniency in them when comes to delivering parcels. There-fore, the Parcel2Go Android Application that has been proposed is to improvise the existing apps in such a way that the dilemma and the inconveniency that the courier services user is facing could be solved.

The snapshot of the activity diagram as shown in Fig. 2. It shows the flow of the application. The application involves three parties such as customer, system and Parcel2Go personnel. Both the customer and personnel will need to use their registered email address to successfully access to the application. After the customer has login to the application, the customer is required to create a new delivery request. After that, they need to enter their delivery destination and the system will help to calculate the quote rate based on the details keyed in.

Subsequently, the customer is able to view a list of quote rate from different courier service provider. If the customer has chosen the Parcel2Go Delivery service, they could view and accept their delivery confirmation and subsequently the system will generate the One Time Pass (OTP) code for the customer. The customer will need to send the OTP code to the receiver. The system then now will update the customer delivery status to "Pending". The personnel now could receive the delivery request and choose whether to accept the delivery request. If the personnel have accepted the request delivery, the system will now then update the customer delivery status to "Accepted by Driver".

Once the personnel click on the "Item Pick Up' button after picking up item from the customer, the system will then again update the delivery status to 'Delivering". After the personnel has delivered the item to the receiver, it will then need to request OTP code from the receiver to be enter into the receiver address page. The system subsequently will verify the OTP code and update the delivery status to "Completed" whilst it could be view by the customer. Meanwhile for the personnel side, it will also prompt a notification showing that the delivery service has been completed and personnel will just need to click on "OK" button.

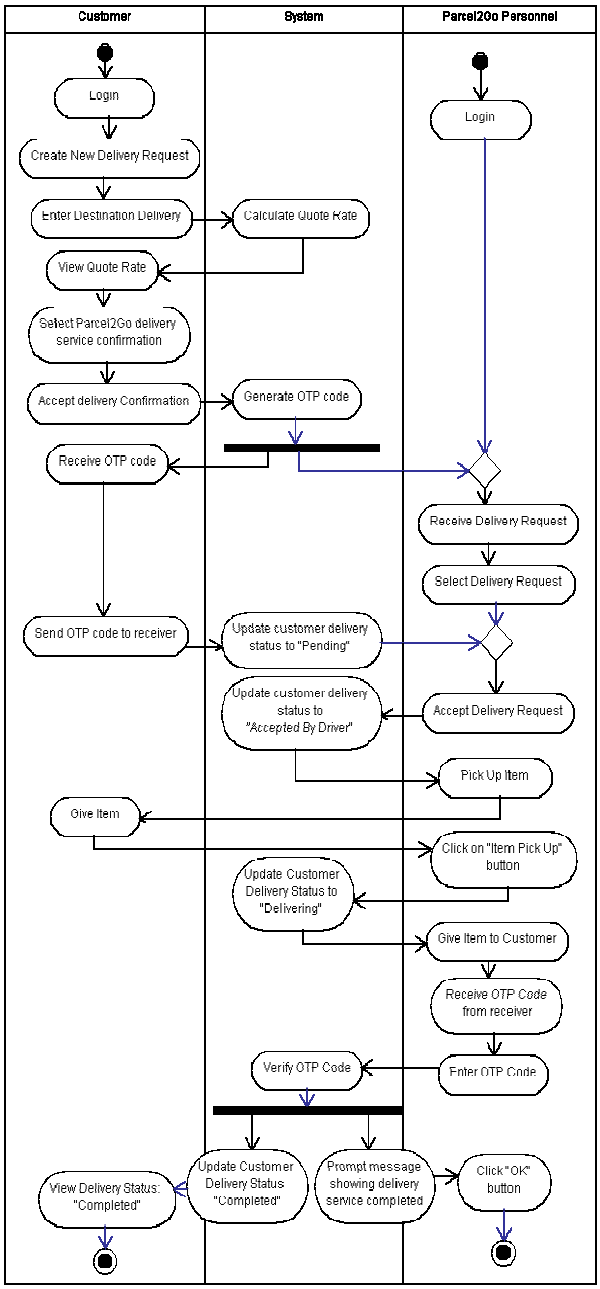


Fig. 2. Parcel2Go Activity Diagram

IV. Result and Discussion

The usability testing has been conducted to 40 respondents of users of the integrated courier service application. The respondents are from various backgrounds such as student, online business seller, professionals, full-time housewives etc. All of the respondents were given a mobile phone that has been installed the Parcel2Go Mobile Application. They are 2 types of user for this mobile application such as customer and personnel. Both the users were asked to complete an online survey regarding their experience after using this application.

The respondents gave positive responses towards the application. Generally, the user interface of Parcel2Go Mobile Application is user friendly and it helps the user to get the estimated quotation of their items that needed to be delivered from different courier services in addition helping those who wants to earn while travelling. The analysis was done gathering the customer's feedback and personnel's feed-back respectively with Likert Scale of five-point scale where used which is 1 signifies strongly disagree by the users and 5 signifies strongly agree by the users.

Fig. 3. has illustrated that all the customers agreed that this application is time saving in obtaining the quotation from different courier services. Besides, majority of them strongly agree that it could solve the inflexibility of courier services by providing door-to-door delivery service to customer and this does not stop them from using this application again when they want to deliver items. As for the reliability of the application, 20 out of 40 customers agree to the reliability of the application despite 6 of them remain neutral due to the lack of trustworthiness. Most of the customer feel that the application is very easy to use despite that some of them is not a tech-savvy person.

The development application is a simulation to show the concept is applicable in an integrated courier services application makes respondents satisfied with the over-all performances. Thus, making the application to achieve its objectives which is to provide convenience for the people who wants to deliver their parcels and the inflexibility of courier services.

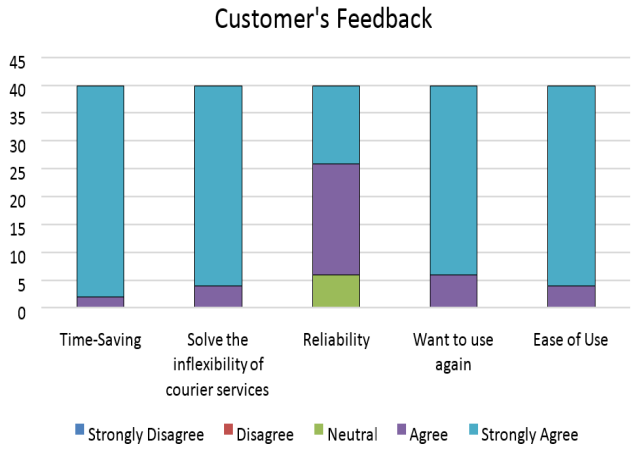


Fig. 3. Customer's Feedback

Based on the Fig. 4., all the personnel strongly agree that they can earn a side in-come while travelling by using this application. Most of the personnel agreed that they have the options to select the customer's delivery request based on their preference. This application concept in attracting more people to help in delivering items was strongly agreed by most of the personnel. Besides, users agreed that the application is very reliable in terms of indicating their delivery task is completed after the receiver gave them an OTP code to be verified by the application whilst it is easy for the personnel to use.

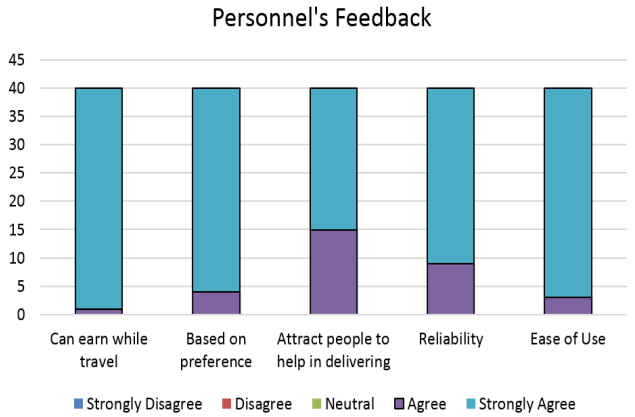


Fig. 4. Personnel's Feedback

The mobile courier service application is needed to make significant impact on the online seller and average people's time and convenience in searching for the best price, service and quality that is provided by the respective courier company. Thus the role of this mobile service application may increase the efficiency and effective-ness of selecting a preferable low cost courier services to carry out the postage delivery.

From the illustrated results, it is apparently indicated that this proposed system, Parcel2Go Android Application, will be a convenient platform for the online business seller and also for the average person who wants to deliver their parcels. As we know that for an average person who at times may have some item to be delivered yet not aware about the cost of courier services charges and the timing that needed to reach to the particular destination. This is because certain courier services the postman will have a fix time to pick up all those items needed to be delivered. Thus once item is dropped in the post office right after the fixed timing that postman come to collect and deliver the items, customers will need to wait another one more working day in order for the parcel to reach to the required destination.

In addition, customers definitely will always opt for a cheaper and faster delivery method, and they will tend to compare the best price rate of certain courier services before deciding which to choose. Subsequently, not all courier services show their direct price rate in their websites as some may go by dynamic rate and this had led to the confusion and time consume in searching for the courier services price information. Certain customers might also need to make a phone call to their customer service to check the price rating if they wants to send in a large bulk. As from the survey, it also

has proven that majority of the courier services users has average spent more than 30 minutes to check out the pricing information.

Therefore, with this proposed Parcel2Go Android Application, it will provide solutions to the loopholes of courier services that the courier users are facing. The solutions whereby this proposed mobile application will includes pricing and timing of delivery of a few domestic courier services player in Malaysia will be shown in the apps for courier users to compare the price rate. Besides Parcel2Go application also provide a door-to-door delivery services which ease user who does not possess a transportation or who wants to save time to be more focus in doing their business whilst save the cost of delivering.

However, keeping critical or high-esteem resources sheltered and secure inside the limits of any business can be a challenge. Ensuring and anchoring similar resources while being transported locally, provincially, the nation over, or around the world displays a completely unique arrangement of issues. mitigates the hazard to the sheltered transport of these basic resources by examining consistence against a typical arrangement of gauges and industry best practices. Defying sophisticated IT environments and a developing rundown of potential system and database security issues, budget-constrained on IT now find it impossible to attend to all known vulnerabilities. Because of the sheer number of patches dispersed and the trouble of measuring the estimation of security fixing to business, mitigating crucial network and application weakness is a constant challenge for any association focused on operational enhancement and security inside the courier industry.

V. CONCLUSION

Courier services applications are widely used in the era of rapid technology advancement. The developed applications, Parcel2Go has become important for people who frequently want their items to be delivered, especially for people who are conducting online businesses. The application be able to generate the quotes of selected courier services in one application and enable a price comparison services. The conducted usability study analyse of respondents' feedbacks shows positive interest. The integrated courier service application implements the concept of service innovation to solve inconvenience of courier service users of getting quotation and the need to travel to post office to drop their items. With the Parcel2Go Mobile Application, the system generate the quotation from different courier services whilst matching up people who needs parcels to deliver from their doorstep to another and people willing to help them and earn while travelling.

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